



CLIMATE



SOSV bets plasma will
change everything from
semiconductors to
spacecraft

Sometimes a tool — like a hammer — comes along and everything starts looking like a nail. But other times, new tools end up being more than a simple blunt-force object.

The investors at [SOSV](#) are betting that plasma will become a nuanced implement capable of everything from enabling fusion power to changing the way semiconductors are made.

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IMAGE CREDITS: LYAGOVY / GETTY IMAGES

It sees so much potential that it plans on investing in more than 25 plasma-related startups over the next five years. It is also opening a new Hax lab space in partnership with the New Jersey Economic Development Authority and the U.S. Department of Energy's Princeton Plasma Physics Laboratory.

[Nuclear fusion](#) is an obvious place to seed plasma startups. The potential power source works by compressing fuel until it turns into a dense plasma, so dense that atoms begin fusing, releasing energy in the process.

“There’s so much here. The best ideas have yet to come to unlock a lot of potential in the fusion

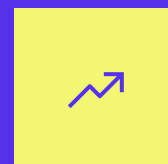
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space,” Duncan Turner, general partner at SOSV, told TechCrunch.

But fusion is just the tip of the iceberg, Turner said. The company has already invested in two companies, including YPlasma, which uses plasma actuators to cool data center chips and control airflow over wind turbine blades.

Finding new uses for plasma in semiconductor manufacturing, where the state of matter already plays a role, could unlock new materials and processes, Turner said. Spacecraft could benefit from plasma thrusters, which are more fuel efficient than traditional thrusters. Harnessing plasma could also unlock [entirely new ways to make chemicals](#) like ammonia for fertilizer or fuel from CO₂.

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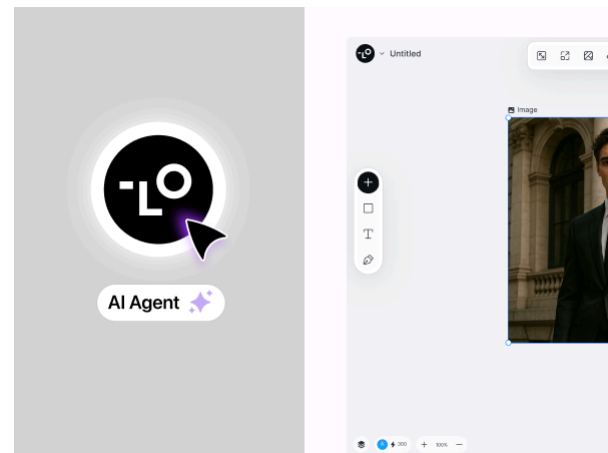
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“All of that combined, we were like, ‘Oh wow, there’s way more than 25 companies here,’” he said. “There’s actually hundreds of companies here.”

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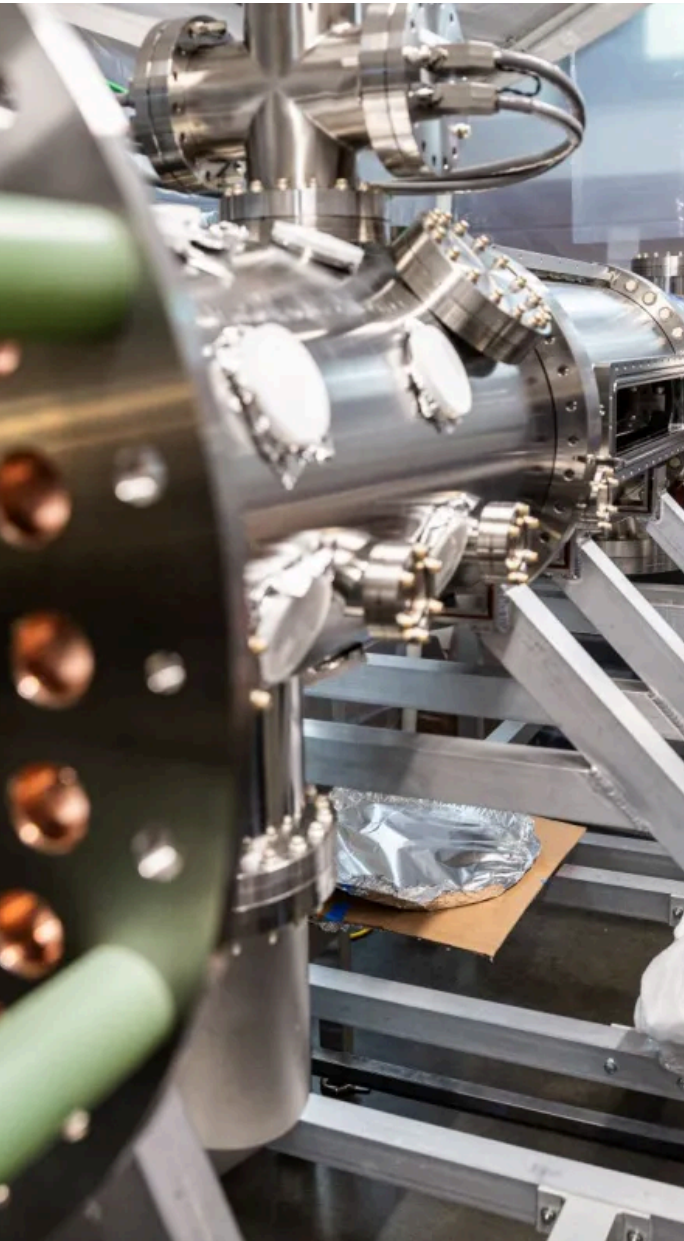


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Zap Energy nets \$160M Series C to advance its lightning-in-a-bottle fusion tech

Tim De Chant — 9:45 AM PDT · June 22, 2022

Fusion startup [Zap Energy](#) has reached two milestones that could nudge it ahead in the race to offer low-cost, carbon-free energy — a \$160 million Series C round and a successful test of a prototype fusion reactor that could pave the way to a commercial version.

[Fusion power](#) has become an unlikely investor favorite as carbon emissions continue to rise and the effects of climate change become more apparent. We have been trying to harness the power of the sun to produce energy for many years, but after tens of billions of dollars and decades of research, fusion remains just out of reach.

Still, clever new approaches to contain scorching hot plasma — which burns at more than 100 million degrees Celsius — have brought fusion power tantalizingly close to reality. Investors are flocking to the field, hopeful that advances wrought by continued research and increasingly sophisticated computer simulations will finally help fusion pull away from its long string of failures.

Zap Energy's oversubscribed Series C was led by Lowercarbon Capital. New investors include Breakthrough Energy Ventures, Shell Ventures, DCVC and Valor Equity Partners. Existing investors Addition, Energy Impact Partners and Chevron Technology Ventures also contributed to the round. The startup's core technology was spun out of research performed at the University of Washington and Lawrence Livermore National Laboratory.

Generally speaking, fusion power generates electricity by fusing hydrogen isotopes (either deuterium or tritium) into helium. The process releases neutrons, which are then captured to generate heat and spin a turbine. Atomic nuclei don't like to fuse, so to coax them close enough for fusion to happen, nuclear scientists use extreme

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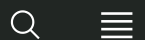
pressure and heat, creating a fourth state of matter known as plasma.

To create and contain the plasma, researchers generally take one of two approaches. In one, they compress hydrogen using [powerful magnetic fields](#) and then send an electric current through the resulting plasma to raise its temperature. In the other, they shoot a fuel pellet with a [powerful laser](#), which both compresses and heats the hydrogen isotopes.

Zap Energy takes a third approach known as a sheared flow stabilized Z-pinch. In the Z-pinch, engineers send a plasma stream



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The electrified plasma in Zap Energy's reactor design is strikingly similar to what happens in a thunderstorm. Lightning bolts superheat the air around them, creating plasma, which is then pinched by the electricity flowing through it. Essentially, Zap Energy is using lightning in a bottle to trigger a fusion reaction.

Researchers have been able to fuse atoms on countless occasions using all three approaches, but the challenge for commercial fusion power has always been designing a reactor that produces more power than is used to run it.

Lasers, magnets and electric fields all require enormous amounts of power. For example, [ITER](#), the massive experimental reactor being built in France, is expected to draw up to 620 MW at its peak — enough to power about 400,000 homes, if briefly.

By using the plasma to create its own magnetic confinement field, Zap's engineers don't have to build costly magnets or lasers, nor do they have to power them, reducing the amount of energy needed to reach breakeven.

Confining plasma, no matter how you go about it, is tricky business, though. Plasma is notoriously unstable under magnetic confinement and that instability is part of what has bedeviled attempts to achieve stable and net-positive fusion power. Early fusion experiments attempted to harness a variation of the Z-pinch but failed because the plasma grew unstable. Those failures drove researchers to largely favor magnetic or laser confinement.

But Zap Energy thinks it has solved the instabilities in the Z-pinch approach — its investors are clearly hoping that’s the case and that Z-pinch reactors will be both cheaper to build and faster to market. The company expects its reactor’s power production will reach the breakeven point within a year.

Fusion power has long been derided as always “just a few years away.” That’s still the case, but if Zap Energy can achieve breakeven and convincingly show that it can push past that, fusion proponents might finally prove their critics wrong.

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World-record fusion experiment produced even more energy than expected

Tim De Chant — 9:45 AM PST · December 13, 2022

It's official: A U.S. Department of Energy lab produced a controlled nuclear fusion reaction that released more energy than it consumed.

On December 5, just after 1:03 a.m. PT, 192 lasers at the National Ignition Facility converged on a small gold cylinder that contained a tiny bead of fuel composed of two isotopes of hydrogen, deuterium and tritium. In a flash, the cylinder vaporized, emitting X-rays that bombarded the fuel pellet, turning its outer diamond layer into an expanding plasma that compressed the fuel inside to the point where its nuclei fused and released a tremendous amount of energy.

IMAGE CREDITS:
CHIP SOMODEVILLA / GETTY IMAGES

How much energy?

When the BB-sized fuel pellet ignited and produced a sustained fusion reaction, it released about 50% more energy than was imparted by the experiment's lasers — the world's largest and most energetic laser system. The lasers' energy heated the fuel pellet to 150 million degrees Celsius and compressed it with a pressure over twice that found at the sun's center.

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In specific terms, the lasers put in 2.05 megajoules of energy and the fusion reaction released 3.15 megajoules. That's even better [than leaked reports suggested](#). The team at the Lawrence Livermore National Laboratory spent the last week deciphering the data to determine the exact results.

Only 4% of the deuterium-tritium fuel burned in the fusion reaction, suggesting plenty of room for improvement. The gold cylinder took about two weeks to manufacture, and the diamond-coated fuel pellet took about seven months to produce.

In scientific and technical terms, the reaction is considered net positive. In real-world terms, it produced far less power than what would be expected of a commercial power plant. To produce the 2.05-megajoule shot, the laser system required 300 megajoules of power, Kim Budil, director of LLNL, said in a press conference today.

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Still, Budil said that the team sees a way to get to “hundreds of megajoules of output” per shot, an amount that would be required for a commercial-scale power plant using the technique, known as inertial confinement.

“We need the private sector to get into the game,” Energy Secretary Jennifer Granholm said.

While a few fusion startups are focused on inertial confinement, the majority are pursuing magnetic confinement, which uses powerful superconducting magnets to corral burning plasma to the temperatures and pressures required for fusion.

The fundamental underlying physics are the same for the two approaches, but the rest of the science and engineering challenges are quite different. In the fusion research world, magnetic confinement is considered low temperature and low pressure, whereas inertial confinement is high temperature and high pressure. Still, there are many places where work on one side benefits the other and vice versa, Carolyn Kuranz, an experimental plasma physicist at the University of Michigan, [told TechCrunch yesterday](#).

The opportunities for shared knowledge have investors heartened, too. “Fusion science has been making steady progress toward energy breakeven on a number of fronts for decades,” said [Phil Larochelle](#), a partner at Breakthrough Energy Ventures. “We expect other approaches to make similar breakthroughs this decade, paving the way for commercial plants in the near future.”

The team had a hunch that last week’s shot would be a good one, though they had no idea just *how* good until the diagnostic data started rolling in. The first thing that principal experimentalist Alex Zylstra did when he saw the initial data was to call a

diagnostic expert to double-check whether it was really as good as he thought.

By the morning, the rest of the team woke up to the promising news, and by the next week, they knew they had achieved a world first.

The shot was the result of refinements put in place after successful experiments in August 2021 and earlier this year. The string of successes was enabled not only by data from previous experiments, but also machine learning trained on that data, Budil said.

When it comes to a timeline for commercializing inertial confinement, Budil remains a realist but is growing increasingly optimistic. Now that the theory has been proven, she sees where specific advancements, from targeting improvements to laser efficiencies, could help bring it to market.

Commercial fusion remains “probably decades away,” she said. “Not six or five as we used to say, but moving into the foreground. A few decades of research into the underlying technologies could put us in the realm of a commercial power plant.”

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